AC Input Protection, Isolated 5 V Supply

File: power.kicad_sch

Zero-Cross Opto-Triac AC Load Control

File: load.kicad_sch

ATtiny13A Controller & ISP

File: mcu.kicad_sch

Radar, Button, LED, Delay Trimmer

File: io.kicad_sch

SPS-24G-01 Smart Presence Sensor Switch - 70 x 56 mm, 2-layer 1.6 mm FR-4, 1 oz, ENIG (SAMAY ELECTRONICS)
Standalone (no WiFi / no BLE). Fits a standard Indian 86 x 86 mm modular wall box.

```
===== HIGH-VOLTAGE (AC) SECTION ====================|===== LOW-VOLTAGE (SELV) SECTION ==========
L in -> F1 T500mA -> ACL_F -+- MOV1 14D471K          |  HLK-PM01 +5V -> ATtiny13A-SU
                           +- CX1 100nF X2 + RB1 1M  |     PB4 <- LD2410B OUT (24 GHz radar)
                           +- NTC1 5D-9 -> HLK-PM01 AC/L  |  PB3 <- 10k delay trimmer (10 s..10 min)
                           +- BTA16-600B MT1         |     PB2 <- override button, PB1 -> status LED
LAMP out <- BTA16 MT2 ; gate <- RG1 180R <- MOC3063   |     PB0 -> MOC3063 LED (opto, SELV side)
snubber RS1 100R + CS1 10nF across MT1-MT2            |  ISP: 6-pin AVR header (program with mains OFF)
```

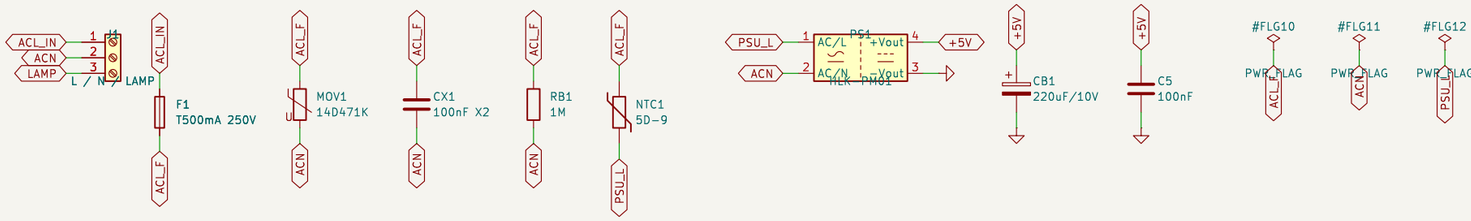
Isolation: HLK-PM01 (reinforced AC-DC) + MOC3063 (7.5 kVrms opto) are the ONLY parts crossing the barrier.
>= 6 mm creepage / >= 3 mm clearance + a routed isolation slot separate the two sections. SEE SAFETY_REVIEW.md.

Top level: block diagram
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Sheet: /
File: bps_switch.kicad_sch

Title: SPS-24G-01 - Top level

Size: A3	Date: 2026-10-24	**Rev: A0**
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DESIGN NOTES

!! MAINS 85–265 VAC on this sheet. All copper left of the HLK-PM01 AC pins is LIVE. !!

Live path: J1.L -> F1 (T500 mA) -> ACL_F. MOV1 (14D471K) + CX1 (100 nF X2) + RB1 (1M bleeder) across L–N.

PSU feed: ACL_F -> NTC1 (5D-9 inrush) -> HLK-PM01 AC/L; AC/N -> Neutral. The lamp inrush does NOT pass NTC1.

HLK-PM01: reinforced-isolation AC-DC, 5 V / 3 W. Its isolation is the primary SELV barrier (>=6 mm creepage).

5 V rail: CB1 220 uF bulk + C5 100 nF. Feeds the radar (~79 mA, bursty). MCU and opto LED.

Layout: AC pins and DC pins of PS1 are 29 mm apart — route the isolation slot between them.

PART NOTES (VERIFY = check against vendor datasheet before release)

J1: Lamp returns to Neutral externally. Mark L/N/LAMP on the lid; wire per the install card.

F1: VERIFY breaking capacity >= prospective fault current; a user-serviceable 5x20 holder is the alt.

MOV1: VERIFY energy rating for the site; a fused MOV or MOV+GDT improves end-of-life safety.

RB1: Bleeder: discharges CX1 when unplugged (< 1 s). Rated for continuous 230 Vrms.

NTC1: Limits turn-on surge into the HLK input cap; the lamp inrush goes through the triac, not here.

PS1: Reinforced AC-DC isolation is the primary SELV barrier. Keep AC and DC pins' copper apart (slot).

Sheet 2: AC Input Protection, Isolated 5 V Supply

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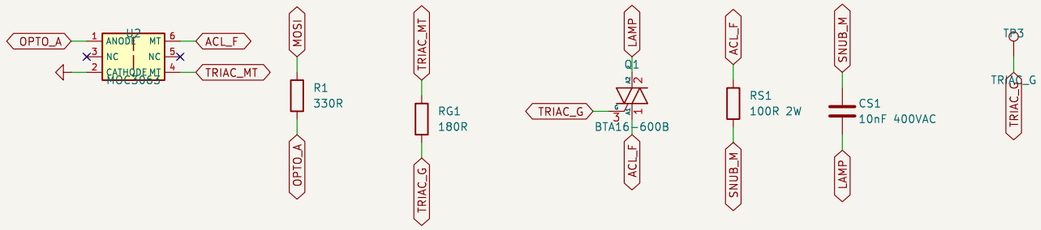
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Sheet: /AC Input Protection, Isolated 5 V Supply/

File: power.kicad_sch

Title: SPS-24G-01 — AC Input Protection, Isolated 5 V Supply

Size: A3	Date: 2026-10-24	**Rev: A0**
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DESIGN NOTES

!! MAINS on the triac (Q1) and the MOC3063 output pins 4/6. The opto is the isolation barrier. !!
MCU PB0/MOSI -> R1 (330R) -> MOC3063 LED (pins 1/2, SELV). Opto output (4/6) drives the BTA16 gate.
MOC3063 is ZERO-CROSSING: it only fires near the mains zero-crossing -> low EMI, low gate inrush.
RG1 180R = gate resistor (ACL_F -> opto -> gate). RS1 100R + CS1 10 nF = dv/dt snubber across MT1-MT2.
Q1 BTA16-600B: MT1(A1)=Line (ACL_F), MT2(A2)=Lamp. Derate / heatsink for lamp currents > ~2 A.
VERIFY per ON AN-3004: peak opto current < 1 A ITSM; 180R OK for a zero-cross opto, raise to 360R if not.

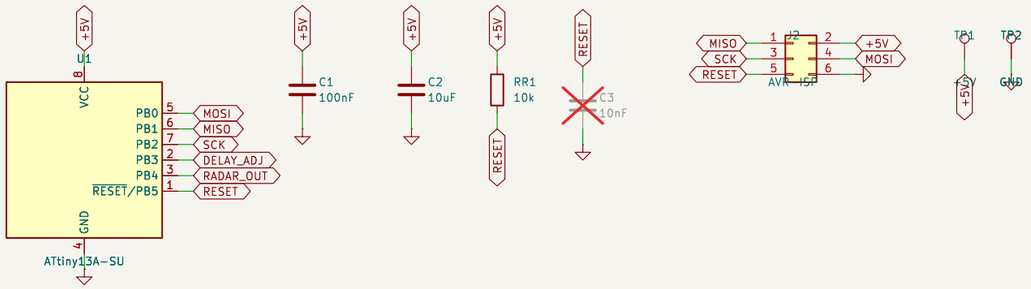
PART NOTES (VERIFY = check against vendor datasheet before release)
U2: Input LED (1,2) = SELV side; output triac (4,6) = mains side. Pins 3,5 NC. 8 mm land spacing keeps the opto's own primary-secondary PCB creepage >= 6 mm.
R1: Opto LED current = (5 - 1.2 V)/330 = 11.5 mA > IFT(10 mA) of MOC3063. Driven by MCU PB0/MOSI.
RG1: 180 ohm per the brief; ON AN-3004 suggests >=360 ohm for 230 V non-ZC -- OK here because MOC3063 is zero-crossing (fires near 0 V). VERIFY peak opto current < ITSM(1 A).
Q1: Non-snubberless part -> external snubber fitted (RS1/CS1). Derate: -6 A continuous no heatsink. VERIFY heatsinking / lamp current; add a clip-on sink for >2 A loads.

Sheet 3: Zero-Cross Opto-Triac AC Load Control
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Sheet: /Zero-Cross Opto-Triac AC Load Control/
File: load.kicad_sch

Title: SPS-24G-01 - Zero-Cross Opto-Triac AC Load Control

Size: A3	Date: 2026-10-24	**Rev: A0**
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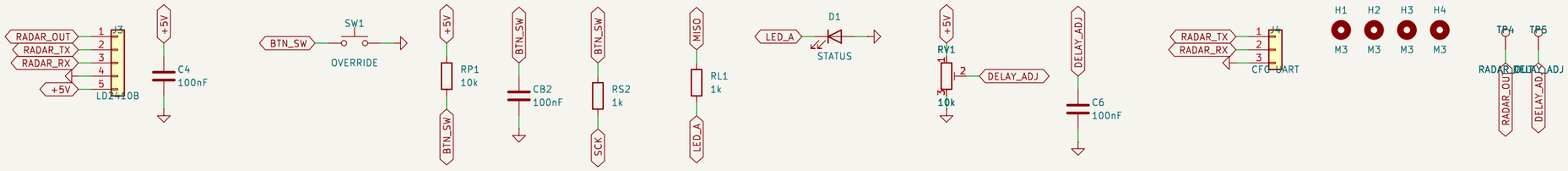


DESIGN NOTES
ATtiny13A-SU at 5 V, internal 9.6 MHz RC. 1 kB flash. Everything on this sheet is SELV (isolated 5 V).
PB5=RESET (ISP), PB3/ADC3=delay trimmer, PB4=radar OUT, PB0=opto drive, PB1=LED, PB2=button.
ISP header J2 is the standard 6-pin AVR pinout. C3 (RESET cap) is DNP so ISP timing is clean.
Program the board on the bench with mains DISCONNECTED (the opto/LED share the ISP lines).

PART NOTES (VERIFY = check against vendor datasheet before release)
U1: Runs at 5 V. Reads LD2410B OUT (3.3 V logic: VIH@5V = 3.0 V, margin ~0.3 V - OK). 9.6 MHz int. RC.
C1: MCU decoupling, close to pin 8
RR1: RESET pull-up (ISP needs RESET released)
C3: DNP: RESET noise cap; keep small or omit so ISP works

Sheet 4: ATtiny13A Controller & ISP
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Sheet: /ATtiny13A Controller & ISP/
File: mcu.kicad_sch
Title: SPS-24G-01 - ATtiny13A Controller & ISP

Size: A3	Date: 2026-10-24	**Rev: A0**
KiCad E.D.A. 9.0.9		Id: 4/5



DESIGN NOTES
LD2410B 24 GHz radar on J3 at the board edge, antenna facing the front cover; no copper under the antenna.
OUT (pin1, 3.3 V) -> PB4: HIGH = person present. UART TX/RX on J4 for the Hi-Link PC config tool (3.3 V).
SW1 override on PB2 (debounced, 1k series so it does not fight the ISP SCK). RP1 10k pull-up, active low.
RV1 10k trimmer -> ADC3 sets the no-one OFF delay (10 s ... 10 min, mapped in firmware).
D1 status LED on PB1: on = presence, slow blink = manual mode, fast blink = fault (e.g. radar not responding).

PART NOTES (VERIFY = check against vendor datasheet before release)
J3: Keep the antenna end clear of copper (no pour under it) and of metal in the lid. OUT high = presence.
C4: local decoupling at the radar header
RP1: button pull-up (active-low)
CB2: debounce; on the switch side, not on SCK
RS2: series into PB2/SCK; isolates the button during ISP
RL1: LED current ~ (5-2)/1k = 3 mA
C6: ADC filter on the wiper

Sheet 5: Radar, Button, LED, Delay Trimmer
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Sheet: /Radar, Button, LED, Delay Trimmer/
File: io.kicad_sch
Title: SPS-24G-01 - Radar, Button, LED, Delay Trimmer
Size: A3 | Date: 2026-10-24 | **Rev: A0**
KiCad E.D.A. 9.0.9 | Id: 5/5